PREDICTIVE ANALYSIS OF PLAYER CHURN
USING MACHINE LEARNING

Student Name: Shane John Mathew
Program: B.Tech Computer Science (AIML)
Institution: TKM Institute of Technology (TKMIT)

PROJECT ABSTRACT

Player retention is a critical metric for sustainability and revenue generation in the gaming industry; however, identifying exactly when a user is likely to abandon a platform remains a complex challenge. The objective of this project was to design and deploy a predictive machine learning pipeline capable of classifying active players at a high risk of churning within a 7-day window. The analysis utilized a behavioral dataset comprising 4,500 player profiles tracked over a 14-day period. Key telemetry extracted for the model included average session lengths, login frequencies, in-app transaction volumes, and frustration indicators such as consecutive failed level attempts and support ticket generation.

The dataset underwent rigorous preprocessing, including the isolation of target variables and the standardization of numerical features using a StandardScaler to ensure algorithmic unbiasedness. A supervised machine learning classification model was then trained to map the relationships between these behavioral inputs and the target churn variable. The model successfully identified underlying behavioral patterns with high predictive accuracy. Feature importance extraction further revealed that "Days Since Last Login" and "Failed Level Attempts" (indicative of player frustration or "rage-quitting") were the primary drivers of churn, whereas higher financial investment via in-app purchases strongly correlated with continued platform loyalty. Ultimately, this predictive architecture provides actionable intelligence, enabling data-driven retention strategies to automatically flag and re-engage at-risk players before they exit the ecosystem.